> FOR SOCIETY

# SOUND SOURCE LOCALIZATION

## MICROPHONE POSSIBILITIES RESEARCH

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# VERSION HISTORY

| Version | Date | Author | Changes |
|---|---|---|---|
| 0.1 | 22-04-2024 | Casey Stijlaart | Layout and basic information and observations. |
| 1.0 | 23-04-2024 | Casey Stijlaart | Completed first version. |

# INTRODUCTION

This document explores the possibilities of using multiple microphones with the Teensy 4.1 development board for sound source localization.

The main objective is to determine the feasibility and practical considerations of connecting and configuring multiple microphones to the Teensy 4.1, with a specific focus on I²S connections and the maximum number of microphones supported.

# TABLE OF CONTENTS

# 1 RESEARCH QUESTIONS

## 1.1 MAIN QUESTION

What are the possibilities and limitations of connecting multiple microphones to the Teensy 4.1 development board for sound source localization applications?

## 1.2 SUB QUESTIONS

- How does the I²S interface facilitate the connection of multiple microphones?
- What is the maximum number of microphones supported by the Teensy 4.1, and how is this determined?

# 2 DEVELOPMENT BOARD

As previously mentioned, the Teensy 4.1 development board is utilized due to its exceptional capabilities in audio processing and versatile hardware features. Besides that, it is equipped with a powerful microcontroller, extensive memory, and a range of connectivity options, the Teensy 4.1 serves as the ideal platform for exploring advanced audio applications such as sound source localization.

The board's specifications boast impressive performance metrics, including high clock speeds, ample memory resources, and dedicated audio processing units. These features not only enable real-time audio processing but also facilitate seamless integration with multiple microphones for precise sound capture and analysis.

For detailed technical specifications and capabilities of the Teensy 4.1 development board, please refer to the official documentation and specifications available at [1].

# 3 MICROPHONE CONNECTIONS

To enable the connection of multiple microphones, the Teensy 4.1 development board utilizes the I²S interface. This interface offers high data transfer rates and synchronization capabilities, making it suitable for audio applications.

[2] states the following : "Teensy 4.0 & 4.1's I²S port has a total of 5 data pins which may each transmit or receive stereo digital audio. This 8-channel input may be used together with the I²S stereo output, but may not be combined with others which use the same physical pins.". This concludes that either 5 mono microphones or 10 stereo microphones can be used with the help of the I²S connections.

The Teensy 4.1 supports up to 10 stereo microphones, with different combinations of data, clock, and word select pins used to achieve this configuration [3].

When connecting 10 microphones, two separate I²S audio objects must be configured, see [2], [4].

## 3.1 CONNECTION OVERVIEW

Up to ten stereo microphones can be connected to the Teensy 4.1 development board. The following table, Table 1, outlines the pin connections required for each microphone.

| Microphone Nr | SD | L/R | WS | SCK |
|---|---|---|---|---|
| 1 | 6 | GND | 21 | 20 |
| 2 | 6 | VCC | 21 | 20 |
| 3 | 8 | GND | 21 | 20 |
| 4 | 8 | VCC | 21 | 20 |
| 5 | 9 | GND | 21 | 20 |
| 6 | 9 | VCC | 21 | 20 |
| 7 | 32 | GND | 21 | 20 |
| 8 | 32 | VCC | 21 | 20 |
| 9 | 5 | GND | 4 | 3 |
| 10 | 5 | VCC | 4 | 3 |

Table 1: Connection Overview stereo microphones and teensy 4.1

## 3.2 CIRCUIT DIAGRAM

The connections between the various pins of the teensy development board and the stereo microphones can be read from the datasheet, [5]. It is good to mention that it is of great importance to have for each pair of microphones, one L/R connected to ground and the other to VDD, as this is a necessity for getting the left and right to work accordingly. For a clear diagram see Figure 1.

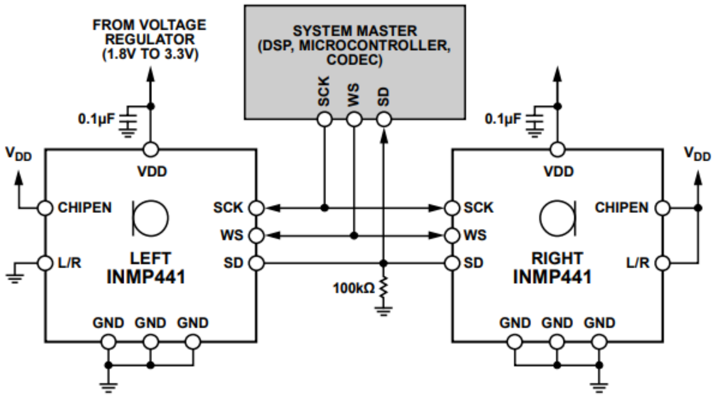


Figure 7. System Block Diagram

Figure 1: INMP441 MEMS Microphone in stereo configuration [5].

When it comes to the test application, this is given in

## 3.3 MICROPHONE CONFIGURATION

When it comes to the configuration for testing the various microphone connections, the following is used for eight microphones, see Listing 1. This example makes use of the AudioInputI2SOct object, which allows for 4 stereo microphone pairs to be connected to one I²S object. Keep in mind, that this is just an example, and the AudioAnalyzePeak objects must be changed accordingly to for the required form of data processing or storing.

```C++
AudioInputI2SOct i2s1;
AudioAnalyzePeak         peak1;
AudioAnalyzePeak         peak2;
AudioAnalyzePeak         peak3;
AudioAnalyzePeak         peak4;
AudioAnalyzePeak         peak5;
AudioAnalyzePeak         peak6;
AudioAnalyzePeak         peak7;
AudioAnalyzePeak         peak8;
AudioConnection          patchCord1(i2s1, 0, peak1, 0);
AudioConnection          patchCord2(i2s1, 1, peak2, 0);
AudioConnection          patchCord3(i2s1, 2, peak3, 0);
AudioConnection          patchCord4(i2s1, 3, peak4, 0);
AudioConnection          patchCord5(i2s1, 4, peak5, 0);
AudioConnection          patchCord6(i2s1, 5, peak6, 0);
AudioConnection          patchCord7(i2s1, 6, peak7, 0);
AudioConnection          patchCord8(i2s1, 7, peak8, 0);
```

Listing 1: eight microphone configuration

For additional microphones beyond the eighth, a separate I²S object is required, AudioInputI2S. This I²S object can only be configured for one stereo microphone pair. The example is given in Listing 2.

```cpp
AudioInputI2S i2s2;
AudioAnalyzePeak         peak9;
AudioAnalyzePeak         peak10;
AudioConnection          patchCord7(i2s2, 8, peak9, 0);
AudioConnection          patchCord8(i2s2, 9, peak10, 0);
```

Listing 2: ten microphone configuration

# 4 POC VALIDATION

For verification that the microphones work as expected, a test setup and application are created. The test is focused on implementing 8 microphones, as this was our set goal. For the connections between the various microphones and the teensy 4.1 development board, see Figure 2. When it comes to the test application, which must be run on the teensy 4.1 development board, this can be found in the given appendix, Section A.

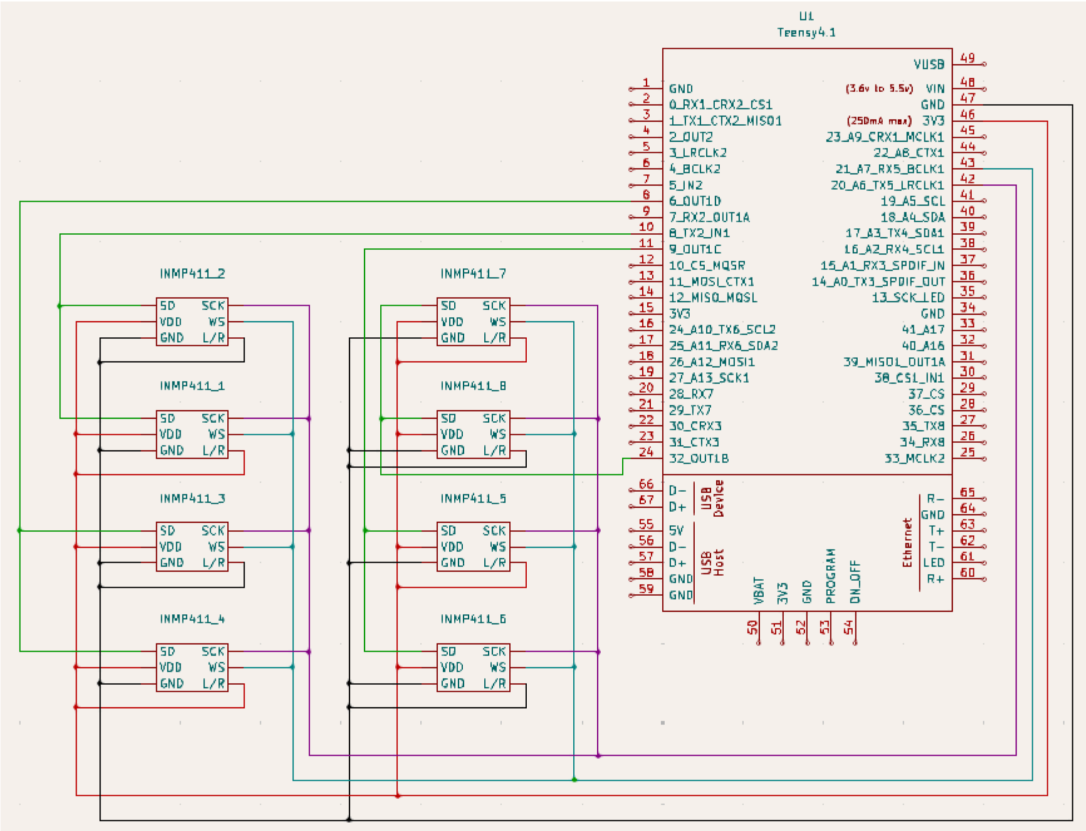


Figure 2: Circuit diagram - Test Setup

Running this test program showcases the capturing of incoming microphone data, where all are similar in value. The slight differences between the values can be explained with the help of each microphone having a slightly different position from the sound source. In the results, it can be seen how generally speaking the amplit values are low, but when a sudden loud noise is detected the amplitude spikes drastically. This can be seen in the following figure, . With the help of these tests, it could be concluded that it is possible to have at least eight microphones connected to a teensy, with the help of just one I²S audio object.

```
Raw Mic 1: 0.003235    Raw Mic 2: 0.003265    Raw Mic 3: 0.003357    Raw Mic 4: 0.003357    Raw Mic 5: 0.003479    Raw Mic 6: 0.001068    Raw Mic 7: 0.003479    Raw Mic 8: 0.001068
Raw Mic 1: 0.001892    Raw Mic 2: 0.001831    Raw Mic 3: 0.002014    Raw Mic 4: 0.001953    Raw Mic 5: 0.001679    Raw Mic 6: 0.000671    Raw Mic 7: 0.001679    Raw Mic 8: 0.000671
Raw Mic 1: 0.002503    Raw Mic 2: 0.002777    Raw Mic 3: 0.002411    Raw Mic 4: 0.002411    Raw Mic 5: 0.002838    Raw Mic 6: 0.000946    Raw Mic 7: 0.002838    Raw Mic 8: 0.000946
Raw Mic 1: 0.002289    Raw Mic 2: 0.002380    Raw Mic 3: 0.002380    Raw Mic 4: 0.002319    Raw Mic 5: 0.002503    Raw Mic 6: 0.000793    Raw Mic 7: 0.002503    Raw Mic 8: 0.000793
Raw Mic 1: 0.001953    Raw Mic 2: 0.001984    Raw Mic 3: 0.001892    Raw Mic 4: 0.001953    Raw Mic 5: 0.002106    Raw Mic 6: 0.000824    Raw Mic 7: 0.002106    Raw Mic 8: 0.000824
Raw Mic 1: 0.004486    Raw Mic 2: 0.004608    Raw Mic 3: 0.004517    Raw Mic 4: 0.004517    Raw Mic 5: 0.004822    Raw Mic 6: 0.001373    Raw Mic 7: 0.004822    Raw Mic 8: 0.001373
Raw Mic 1: 0.002441    Raw Mic 2: 0.002594    Raw Mic 3: 0.002350    Raw Mic 4: 0.002380    Raw Mic 5: 0.002472    Raw Mic 6: 0.000702    Raw Mic 7: 0.002472    Raw Mic 8: 0.000702
Raw Mic 1: 0.752068    Raw Mic 2: 0.760430    Raw Mic 3: 0.777490    Raw Mic 4: 0.767571    Raw Mic 5: 0.847774    Raw Mic 6: 0.459304    Raw Mic 7: 0.847774    Raw Mic 8: 0.459304
Raw Mic 1: 0.066103    Raw Mic 2: 0.039369    Raw Mic 3: 0.071078    Raw Mic 4: 0.075167    Raw Mic 5: 0.036531    Raw Mic 6: 0.002380    Raw Mic 7: 0.036531    Raw Mic 8: 0.002380
Raw Mic 1: 0.002625    Raw Mic 2: 0.002747    Raw Mic 3: 0.002533    Raw Mic 4: 0.002625    Raw Mic 5: 0.002747    Raw Mic 6: 0.000732    Raw Mic 7: 0.002747    Raw Mic 8: 0.000732
Raw Mic 1: 0.002564    Raw Mic 2: 0.002472    Raw Mic 3: 0.002503    Raw Mic 4: 0.002625    Raw Mic 5: 0.002503    Raw Mic 6: 0.000732    Raw Mic 7: 0.002503    Raw Mic 8: 0.000732
Raw Mic 1: 0.003754    Raw Mic 2: 0.003784    Raw Mic 3: 0.003723    Raw Mic 4: 0.003754    Raw Mic 5: 0.003937    Raw Mic 6: 0.000916    Raw Mic 7: 0.003937    Raw Mic 8: 0.000916
Raw Mic 1: 0.004425    Raw Mic 2: 0.004212    Raw Mic 3: 0.004700    Raw Mic 4: 0.004517    Raw Mic 5: 0.003632    Raw Mic 6: 0.001038    Raw Mic 7: 0.003632    Raw Mic 8: 0.001038
Raw Mic 1: 0.002503    Raw Mic 2: 0.002319    Raw Mic 3: 0.002655    Raw Mic 4: 0.002594    Raw Mic 5: 0.002228    Raw Mic 6: 0.000671    Raw Mic 7: 0.002228    Raw Mic 8: 0.000671
Raw Mic 1: 0.002380    Raw Mic 2: 0.002319    Raw Mic 3: 0.002350    Raw Mic 4: 0.002258    Raw Mic 5: 0.002136    Raw Mic 6: 0.000641    Raw Mic 7: 0.002136    Raw Mic 8: 0.000641
```

Figure 3: Results of the test setup.

# 5 CONCLUSION

The Teensy 4.1 development board offers significant flexibility for connecting multiple stereo microphones, with support for up to 10 microphones. While this capability opens up possibilities for sound source localization applications, careful consideration of wiring configuration and proper testing and validation are essential. For the project implementation, we will utilize eight microphones distributed across data pins 6, 8, 9, and 32 to efficiently achieve the objectives.

# GLOSSARY

. 4

***AudioAnalyzePeak***: Refers to an audio analysis function of Teensy used for detecting peaks or amplitude levels in audio signals. 4

***AudioInputI2S***: Specific terminology related to the singular or dual audio input configuration for the Teensy development board. 5

***AudioInputI2SOct***: Specific terminology related to the 8 channel audio input configuration for the Teensy 4.0 and 4.1 development board. 4

***I²S***: Inter-IC Sound, or I²S, is an electrical serial bus interface standard used for connecting digital audio devices. II, 1, 3, 4, 5, 6

# REFERENCES


[1] "Teensy 4.1 Specs." [Online]. Available: https://www.pjrc.com/store/teensy41.html

[2] "Teensy Audio Tool I2S OCT." [Online]. Available: https://www.pjrc.com/teensy/gui/?info=AudioInputI2SOct

[3] "Teensy I2S Forum." [Online]. Available: https://forum.pjrc.com/index.php?threads/teensy-4-1-with-multiple-i2s-microphones.73632/

[4] "Teensy Audio Tool I2S." [Online]. Available: https://www.pjrc.com/teensy/gui/?info=AudioInputI2S

[5] InvenSense, "INMP441," no. DS-INMP441-00. 2014. [Online]. Available: https://www.invensense.com/wp-content/uploads/2015/02/INMP441.pdf

# A TEST-PROGRAM.CPP

```cpp
/**
* @file test-program.cpp
* @author Casey Stijlaart
* @brief Test program for veryifying 8 microphones connected to the teensy 4.1
* @version 0.1
* @date 2024-04-22
*
* @copyright Copyright (c) 2024
*
*/

#include <Arduino.h>
#include <Audio.h>
#include <SD.h>
#include <SPI.h>
#include <SerialFlash.h>
#include <Wire.h>

AudioInputI2SOct i2s1;
AudioAnalyzePeak peak[8] = {AudioAnalyzePeak(), AudioAnalyzePeak(),
                            AudioAnalyzePeak(), AudioAnalyzePeak(),
                            AudioAnalyzePeak(), AudioAnalyzePeak(),
                            AudioAnalyzePeak(), AudioAnalyzePeak()};
AudioConnection patchCord[8] = {
    AudioConnection(i2s1, 0, peak[0], 0), AudioConnection(i2s1, 1, peak[1], 0),
    AudioConnection(i2s1, 2, peak[2], 0), AudioConnection(i2s1, 3, peak[3], 0),
    AudioConnection(i2s1, 4, peak[4], 0), AudioConnection(i2s1, 5, peak[5], 0),
    AudioConnection(i2s1, 6, peak[6], 0), AudioConnection(i2s1, 7, peak[7], 0)};

void setup() {
  Serial.begin(9600);
  AudioMemory(12);
}

void loop() {
  if (peak[0].available() && peak[1].available() && peak[2].available() &&
      peak[3].available() && peak[4].available() && peak[5].available() &&
      peak[6].available() && peak[7].available()) {
    for (int i = 0; i < 8; i++) {
      float amplitude = peak[i].read();
      Serial.print("Raw Mic ");
      Serial.print(i + 1);
      Serial.print(": ");
      Serial.print(amplitude, 6);
      Serial.print("\t");
    }
```

```
        Serial.println();
      }
      delay(300);
    }
```